Mohammad Akbar Bin Shah

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Education

Kansas State University — Manhattan, KS, USA
Ph.D. in Computer Science (GPA: 4.00/4.00) · *Advisor: Dr. Doina Caragea* — *Aug. 2025 – Present*

Chittagong University of Engineering & Technology (CUET) — Chattogram, Bangladesh
B.Sc. in Computer Science & Engineering (CGPA: 3.71/4.00, ranked 14th of 132) — *Feb. 2019 – May 2024*

- Government Technical Scholarship in all 8 semesters.

Research Experience

Graduate Research Assistant — Jan. 2026 – Present
Kansas State University · NSF SaTC 2.0 project (Grant No. 2523438) — *Manhattan, KS, USA*

- **Infant distress recognition (ICTAI 2026, first author):** Built a **PyTorch** pipeline that fine-tunes **OpenFace-3.0** on infant faces and fuses it with **PyAFAR** action units or **MediaPipe** landmark geometry, reaching **0.91 weighted F1** on 19 unseen hospital patients vs. 0.73 unadapted; validated with subject-level bootstrap and ablations, and used **Grad-CAM** and **SHAP** to show the model attends to the facial regions clinicians use to score infant pain.
- **(Ongoing) Real-time tracheostomy-tube tracking:** Implementing Meta's **SAM 2/3** and **DINOv2** models for fully automatic, real-time tracking of an infant's tracheostomy tube to flag dislodgement.
- **(Ongoing) 3D bedside reconstruction:** Implementing Meta's **SAM 3D** models for periodic 3D reconstruction of the infant's body and surrounding objects, toward complete bedside awareness.

Undergraduate Researcher — 2023 – May 2024
Chittagong University of Engineering & Technology · Supervisor: Dr. Mahfuzulhoq Chowdhury — *Chattogram, Bangladesh*

- **Federated diabetic retinopathy grading (SN Computer Science, first author):** Implemented **Swin Transformer** federated learning with **FedAvg** and **FedProx** for five-stage classification across three heterogeneous datasets.
- Collected a private patient dataset from Bangladesh Eye Hospital to test on an unseen site; reached **76.8%** accuracy (3.7% above prior work) and **78.0%** on the unseen hospital's patients.

Publications

- **M. A. B. Shah**, A. Zareian, V. Bondalakunta, N. Haque, S. Wei, X. Wan, M. Dunham, G. Amariucai, D. Caragea. "Multi-Representation Facial Feature Fusion for Cross-Domain Infant Distress Recognition." *IEEE ICTAI*, 2026 (accepted).
  Infant-specific adaptation drives cross-hospital transfer; fusing a second facial representation adds a further gain.
- V. Bondalakunta, A. Zareian Jahromi, **M. A. B. Shah**, N. Haque, G. T. Amariucai, D. Caragea, S. Wei, X. Wan, M. E. Dunham. "A Privacy-Preserving Framework for Sharing and Learning from Scarce Medical Video." *HealthSec Workshop*, 2026 (accepted).
  Face-swap anonymization so hospitals can share patient video; off-the-shelf swappers still leak identity.
- A. Zareian Jahromi, V. Bondalakunta, **M. A. B. Shah**, N. Haque, S. Wei, G. T. Amariucai. "Picture the Epsilon: Pursuing Identity-Level Privacy Guarantees for Images." *arXiv:2608.17147*, 2026.
  Compares four identity-level privacy audits for face generators; only a held-out classifier test is valid.
- **M. A. B. Shah**, T. Azhar, Asaduzzaman, M. Chowdhury. "A Swin Transformer-Based Federated Learning Approach for Classification of Diabetic Retinopathy." *SN Computer Science*, 2025.
  Grades diabetic retinopathy across hospitals without sharing patient images; tested on an unseen hospital.

Teaching and Industry Experience

Graduate Teaching Assistant — Aug. 2025 – Dec. 2025
Kansas State University · CIS 115 Introduction to Computing Science — *Manhattan, KS, USA*

- Helped students with computational thinking, algorithm design, introductory programming and Boolean logic.

Lecturer — Jan. 2025 – Jul. 2025
East Delta University — *Chattogram, Bangladesh*

- Taught Data Structures, Computer Networks, and Microprocessor & Interfacing; assisted in Integral Calculus, Vector Analysis & Linear Algebra.

Industrial Attachment — Sep. 2023 – Oct. 2023
Upay (UCB Fintech Company Limited) — *Dhaka, Bangladesh*

- Co-developed a credit-scoring system that predicts Upay users' loan eligibility from transaction history and customer data using **multinomial logistic regression**, served through a **React** and **Django** web interface.

Technical Skills

ML Foundations: Deep learning, computer vision, domain adaptation, self-supervised learning, federated learning
Frameworks & Libraries: PyTorch, OpenFace-3.0, MediaPipe, PyAFAR
Languages: Python, SQL (MySQL, PostgreSQL), C, C++